



DELHI PUBLIC SCHOOL NEWTOWN
SESSION 2022-2023
HALF YEARLY EXAMINATION

CLASS : X

FULL MARKS : 100

SUBJECT : ENVIRONMENTAL APPLICATIONS

TIME : 2 HOURS

[SET-B]

Answers to this Paper must be written on the paper provided separately.

You will not be allowed to write during the first 15 minutes.

This time is to be spent in reading the Question Paper.

The time given at the head of this Paper is the time allowed for writing the answers.

Attempt all questions from Section A and any four questions from Section B
The intended marks for questions or parts of questions are given in brackets [].

This question paper consists of five printed pages.

SECTION A (40 Marks)

Attempt all questions from this Section

Question 1

Choose the correct answer and write the correct option.

[20]

- i. Curitiba is famous for
 - a) Subways
 - b) Metro services
 - c) Public transport
 - d) Private transport
- ii. Which of the following is not true about deforestation?
 - a) Population explosion is one of the reasons for deforestation.
 - b) Clearing of forest for agriculture causes deforestation.
 - c) Deforestation is taking place only in developing countries.
 - d) Cash crop economy of third world is a cause of deforestation.
- iii. Which of the following is not a viable protection against deforestation?
 - a) Reduce the consumption of forest and related products
 - b) Boycott products of companies involved in deforestation
 - c) Privatisation of forest land
 - d) Environmental education
- iv. What is called for the method of removing dust and other particles from water using filters?
 - a) Filtration
 - b) Sublimation
 - c) Rainwater harvesting
 - d) Evaporation

- v. Why is it difficult to recycle plastics?
 - a) It is very hard
 - b) It comes in different sizes
 - c) It is adhesive
 - d) It contains different types of polymer resins
- vi. Why is recycled paper banned for use in food containers?
 - a) Because it creates a lot of spaces
 - b) Because it creates contamination
 - c) Because paper can be used only one time
 - d) Because paper is very thick and can't cover the food containers
- vii. The process of burning municipal solid wastes under suitable temperature and conditions in a specific furnace is called ____.
 - a) Landfill
 - b) Incineration
 - c) Recycling
 - d) Vermicomposting
- viii. Which of the following statements is true about zero waste management?
 - a) Separate collection of each kind of waste
 - b) Segregation of garbage at the source
 - c) Community involvement
 - d) All of the above
- ix. Water is a renewable resource because
 - a) it is being recycled by human beings.
 - b) it is renewed and recharged through hydrological cycle.
 - c) it is being renewed through reduction.
 - d) it can be reused again.
- x. In semi-arid regions of Rajasthan the traditional system of storing drinking water in underground tanks are called
 - a) Dugwells
 - b) Johads
 - c) Tankas
 - d) None of the above
- xi. Extensive planting of trees to increase cover is called
 - a) Afforestation
 - b) Agroforestation
 - c) Deforestation
 - d) Social forestry
- xii. The major cause for land degradation in our country is
 - a) Soil erosion
 - b) Pollution of soil
 - c) Water-logging
 - d) None of the above
- xiii. Which of the following is not an advantage of CNG?
 - a) Reduces air pollution
 - b) Conserves fossil fuel
 - c) Easily available in rural areas

- d) Cheaper
- xiv. Check dams or embankments built in Rajasthan to collect rainwater during monsoon are called
 - a) Baoli
 - b) Temple pond
 - c) Johad
 - d) Eris
- xv. Which of the following has the aim of conservation of wetlands?
 - a) Kyoto Protocol
 - b) Ramsar Convention
 - c) Basel convention
 - d) Geneva Convention
- xvi. The main stake holder of community forestry are
 - a) State government
 - b) Central government
 - c) Local people
 - d) Forest officials
- xvii. Activated sludge is for treating
 - a) Toxic gases
 - b) Waste water
 - c) Nuclear waste
 - d) Municipal solid waste
- xviii. Which one of the following is NOT a wet waste?
 - a) Banana peel
 - b) Leaves
 - c) Egg shells
 - d) Rubber
- xix. Find the odd one out
 - a) Alcohol
 - b) Vegetable oil
 - c) Liquid hydrogen
 - d) Glycerine
- xx. When plantation activity takes place in an area where trees have been uprooted in a cyclone is known as
 - a) Afforestation
 - b) Reforestation
 - c) Deforestation
 - d) Agroforestation

Question 2

- a) Define soil management. [2]
- b) Contrast between tree breaks and check dams. [2]
- c) Suggest ways to protect heritage sites like Taj from air pollution. [3]
- d) Alternative fuels can be a disadvantage-Justify [3]

Question 3

- a) Why are radioactive wastes harmful? [2]
- b) What is the impact of fuel wood burning on village women? [2]
- c) How did TBS contribute for soil conservation? [3]
- d) What is JFM? Who is the father of JFM? Who are the parties involved in JFM. [3]

SECTION B (60 MARKS)

(Attempt any four questions from this section)

Question 4

- a) You went for an excursion in a mountainous region and saw lack of vegetation and greenery there. What can be the possible reasons for that? [5]
- b) What initiatives can you take to increase the tree cover of your locality? [5]
- c) Write about any five applications of remote sensing. [5]

Question 5

- a) Draw and explain the soil profile. [5]
- b) What is Bharat Norm? Write the advantages and the limitations of it. [5]
- c) Eutrophication leads to biomagnifications... give your views on the statement with valid reasons. [5]

Question 6

- a) Today's world is facing water crisis. Suggest any five ways to mitigate the problem. [5]
- b) How can economic policy measures help to control vehicular pollution? [5]
- c) Which city is known as the 'city of dawn' and why? [5]

Question 7

- a) Discuss the problems of waste disposal in India. [5]
- b) Name the pollution control device which separates pollutants by using the following principles:
 - Electrical charge
 - Centrifugal force
 - High temperature combustion
 - Basic sprayWhat is the method known as which uses all these devices to control pollution? [5]
- c) CNG has many advantages as a fuel, still it is not much in use compared to fossil

fuels. Give valid reasons supporting the statement. [5]

Question 8

- a) Suggest any five ways of combating deforestation. [5]**
- b) Enumerate the soil conservation strategies. [5]**
- c) A large variety of alternative fuels and additives have evolved as a choice to be used in vehicles. Discuss. [5]**

Question 9

- a) What is meant by waste water treatment? What is the objective of the process? What are the different steps of the process? [5]**
- b) What are the different techniques of watershed management? [5]**
- c) You have learnt about various water harvesting techniques. Name and describe the one you would want to implement in your locality. [5]**